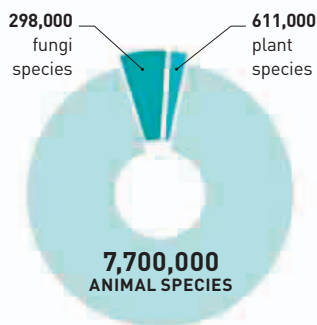




These fragments of comet Shoemaker-Levy 9 were photographed by the Hubble Space Telescope two months before their collisions with Jupiter. Shoemaker-Levy 9 was the first comet ever observed in orbit around anything other than the Sun.



Biodiversity

More than eight million plant, animal, and fungal species are known—there are many more undiscovered.

Japanese company Hitachi developed a new technology, which was called in-plane switching, making it possible to use large LCDs as television screens. By 2007, LCD televisions were outselling those with cathode ray tube screens.

In 1993, the **UN Convention on Biological Diversity** came into force. This treaty, which opened for signatures at the Earth Summit, aims to **conserve biodiversity** and encourage the sharing of profits made utilizing traditional knowledge.

The British company Blatchford unveiled their first prosthesis that was **controlled by a microchip**. This knee prosthesis could automatically adjust to the wearer's gait and became commercially available in 1993.

EIGHT YEARS AFTER THE FIRST FIELD TRIALS of genetically modified crops (see 1986), the first **commercially available genetically modified food** became available this year, when American company Calgene's Flavr Savr tomato was **approved for sale** by the US Food and Drug Administration. Calgene had inserted a gene into the genome of the tomato that hindered the production of an enzyme normally involved in breaking down of cell walls and softening of the fruit. The result was a tomato that could **stay firm and fresh for longer**. The Flavr Savr was discontinued in 1997, after initial commercial success waned.

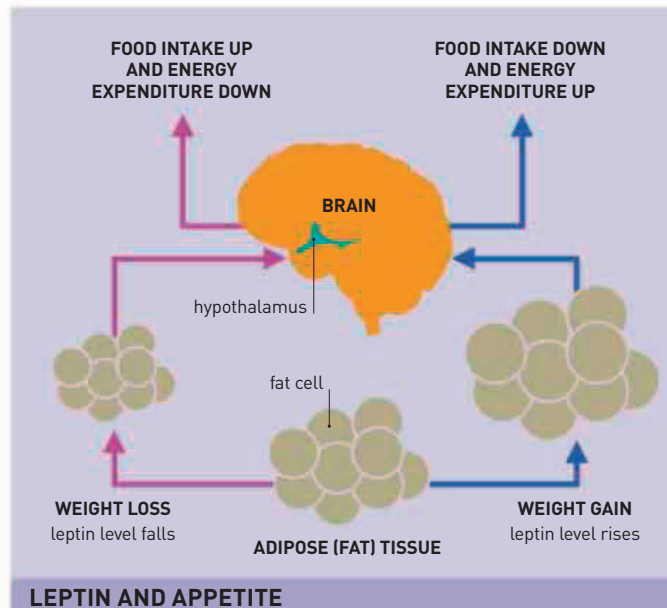
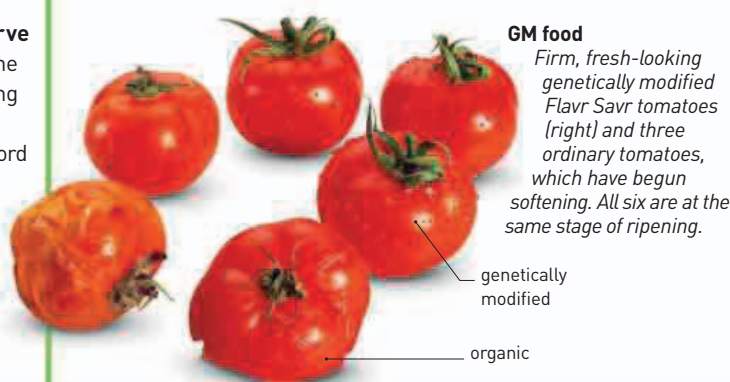
Astronomers around the world had their telescopes fixed on Jupiter in July, when 21 mountain-sized fragments of **comet Shoemaker-Levy 9** plunged into the atmosphere

of the planet. The comet had been captured by Jupiter's gravitational influence, and had probably been in orbit around the planet for more than 20 years. It was discovered in 1993, by American astronomers **Eugene Shoemaker** (1928–97), **Carolyn Shoemaker** (b.1929), and Canadian astronomer **David Levy** (b.1948). The comet had broken into fragments after passing close to the huge planet some time in 1992, and the pieces formed a chain about 6,000 miles (10,000 km) long. The impacts produced explosions that left scars in the atmosphere. The **Galileo space probe** (see 1995), en route to Jupiter, was well placed to gather data and images of the collisions.

In December, a team led by American medical researcher Jeffrey Friedman (b.1954) reported the discovery of a hormone that is **closely involved**

with appetite and therefore with obesity. The hormone, named leptin from the Greek word leptos, meaning thin, **acts on the brain's hunger center** in the hypothalamus (see panel, below). The discovery was made after Friedman's team studied mice with a mutant gene that gave them a voracious appetite. The mutant obese mice were first

discovered in 1950. Injecting leptin into the blood of the obese mice caused the mice to **eat much less and lose weight rapidly**. Medical researchers hoped that leptin might form the basis of a cure for morbid obesity in humans, but it remains an elusive hope.



The energy-regulating hormone leptin is produced by adipocytes (fat cells). Leptin levels are controlled by an area of the brain called the hypothalamus. A person gains weight when the adipocytes in their body fat store more fat. In this state, adipocytes produce more leptin, and the hypothalamus reduces appetite, leading to weight loss. If person's weight reduces, leptin level falls, and appetite increases.

December 29, 1993
The Convention on Biological Diversity comes into force

March 21 The UN Framework Convention on Climate Change comes into force

March 14 Linux 1.0.0 released—the first full version of the popular open-source computer operating system

May 21 The Flavr Savr tomato becomes the first commercially available genetically modified food

July Large fragments of Comet Shoemaker-Levy 9 collide with Jupiter

July German company Fraunhofer releases software for encoding digital audio in the mp3 format

December 1 Discovery of leptin, a hormone released by fat cells that controls appetite